Date: 11/06/2025

Lab Practical #02:
Study of different network devices in detail.

Practical Assignment #02:
1. Give difference between below network devices.
   - Hub and Switch
   - Switch and Router
   - Router and Gateway
2. Working of below network devices:
   - Repeater
   - Modem (DSL and ADSL)
   - Hub
   - Bridge
   - Switch
   - Router
   - Gateway

Hub and Switch

No.	Hub	Switch
1	Operates at OSI Layer 1 (Physical Layer).	Operates at OSI Layer 2 (Data Link Layer).
2	Broadcasts data to all connected devices.	Sends data only to the destination MAC address.
3	Cannot filter or manage traffic.	Can filter traffic using MAC addresses.
4	Creates more collisions in network.	Reduces collisions using separate collision domains.
5	Slower and outdated.	Faster and more efficient for modern networks.

Switch and Router

No.	Switch	Router
1	Operates at OSI Layer 2 (Data Link Layer).	Operates at OSI Layer 3 (Network Layer).
2	Connects devices within a local network (LAN).	Connects multiple networks (LAN to WAN or LAN to LAN).
3	Uses MAC address for data forwarding.	Uses IP address for data forwarding.
4	No NAT (Network Address Translation).	Performs NAT to route traffic between networks.
5	Does not assign IP addresses.	Can assign IP addresses using DHCP (Dynamic Host Configuration Protocol).

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Router and Gateway

No.	Router	Gateway
1	Operates at OSI Layer 3 (Network Layer).	Can operate across multiple OSI layers.
2	Connects similar networks (e.g., LAN to Internet).	Connects dissimilar networks (e.g., TCP/IP to other protocols).
3	Uses IP routing to forward packets.	Translates between protocols as needed.
4	Common in homes and offices.	Common at network edge or enterprise systems.
5	Routes packets between IP-based networks.	Acts as a protocol converter.

Working of below network devices:

1. Repeater
   - A repeater regenerates and amplifies the signal in a network to extend the transmission distance. It operates at the Physical Layer (Layer 1) and does not understand data — it simply boosts the signal to avoid degradation.
2. Modem (DSL and ADSL)
   - A modem (modulator-demodulator) converts digital signals from a computer to analog signals for telephone lines and vice versa.
      i. DSL (Digital Subscriber Line) uses phone lines for internet without interfering with voice calls.
      ii. ADSL (Asymmetric DSL) provides higher download speeds than upload speeds, suitable for home use.
3. Hub
   - A hub is a basic networking device that operates at the Physical Layer (Layer 1). It simply repeats incoming signals to all ports, causing network collisions and reduced efficiency. Hubs are rarely used in modern networks.
4. Bridge
   - A bridge operates at the Data Link Layer (Layer 2). It connects two or more network segments and filters traffic based on MAC addresses. It helps in segmenting networks to reduce collisions and improve performance.
5. Switch
   - A switch operates at the Data Link Layer (Layer 2) of the OSI model. It maintains a MAC address table to intelligently forward frames only to the intended recipient port. This reduces unnecessary data traffic and collisions. Switches can also work at Layer 3 with routing capabilities (Layer 3 switch).
6. Router
   - A router operates at the Network Layer (Layer 3). It determines the best path for data packets to travel from source to destination across different networks. It uses IP addresses to make routing decisions and performs NAT (Network Address Translation) and DHCP (Dynamic Host Configuration Protocol).

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7. Gateway
   - A gateway connects networks that use different protocols. It acts as a translator that allows communication between dissimilar systems. Gateways often perform data format conversions and can operate at various OSI layers depending on the function (e.g., email, VoIP, web gateways).